

Compilers' Theory

Lecture 4

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HOPCROFT ALGORITHM
DFA MINIMIZATION

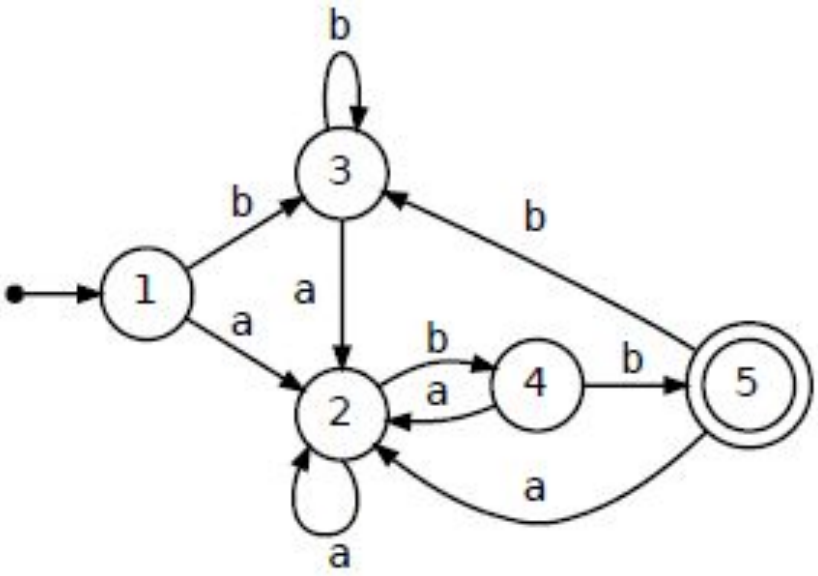
DFA Minimization Theory (Hopcroft's Algorithm)



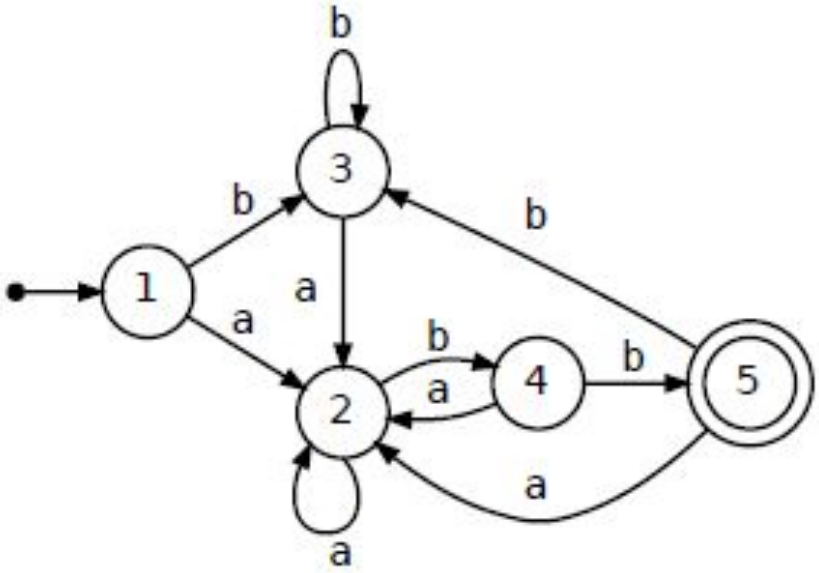
- **Why Minimize?** Subset construction can yield redundant states. Minimization shrinks the state count to reduce the memory footprint and keep transition tables cache-friendly.
- **Core Strategy (Optimistic Partitioning):**
  - Start by splitting states into two initial super-states:
    - Non-Accepting ($\mathbf{T_0}$) and Accepting ($\mathbf{T_1}$).
  - For each super-state $\mathbf{T_i}$ and each character **c**:
    - If states in $\mathbf{T_i}$ transition to different super-states on **c**, then $\mathbf{T_i}$ is inconsistent.
    - Split $\mathbf{T_i}$ into consistent smaller sub-groups.
  - Repeat until no more splits are possible.

Example

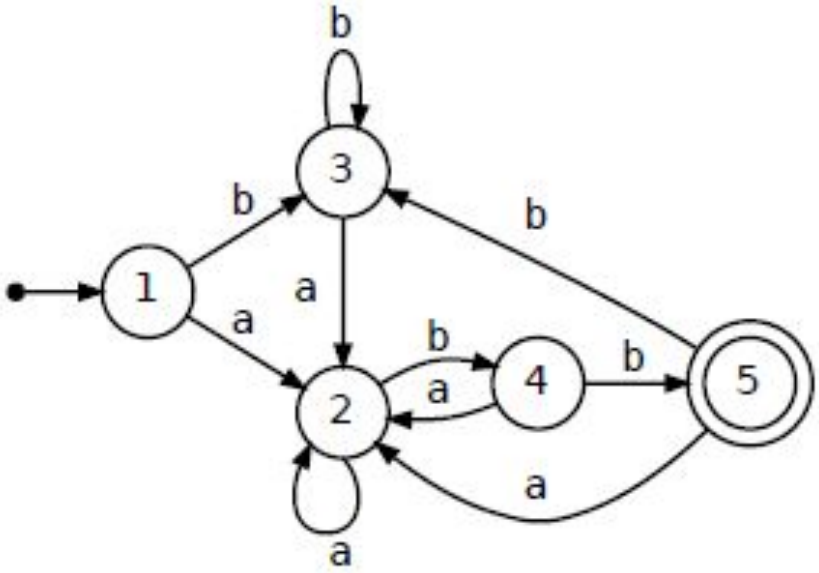
- Suppose we have the following non-optimized DFA and wish to reduce it to a smaller DFA:



q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3

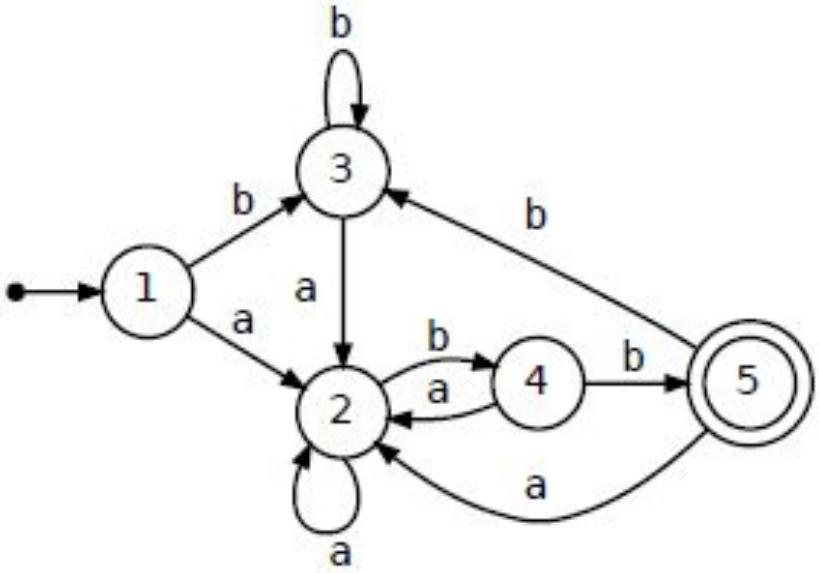


q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}

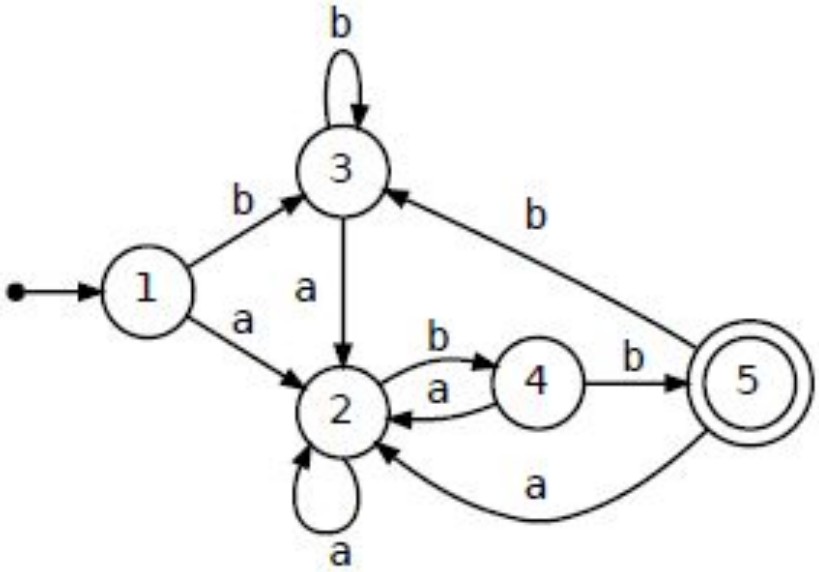
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⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}

Step 2:

q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
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4	2	5
5*	2	3

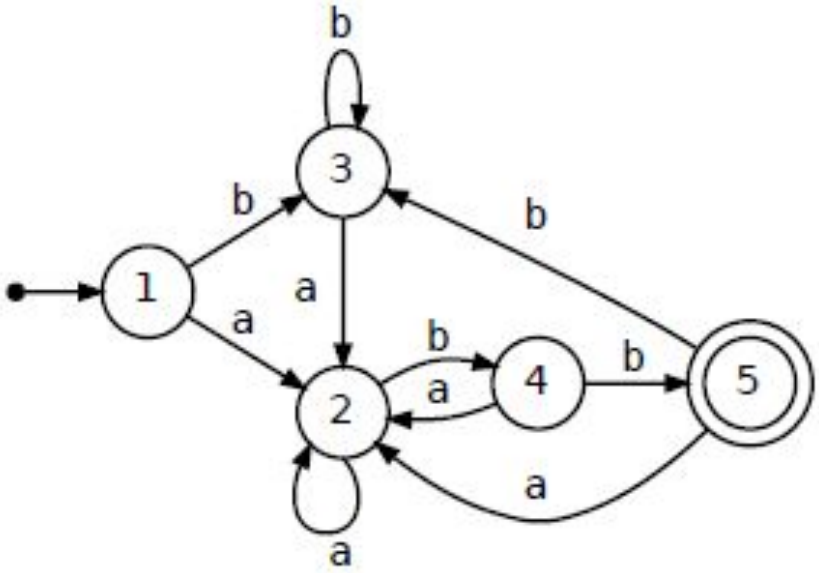


Step 1: **{1,2,3,4}** {5}

Step 2:

{1,2}
{1,3}
{1,4}
{2,3}
{2,4}
{3,4}

q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: **{1,2,3,4}** {5}

Step 2:

{1,2} => **{{~~2,2~~},{3,4}}**

{1,3}

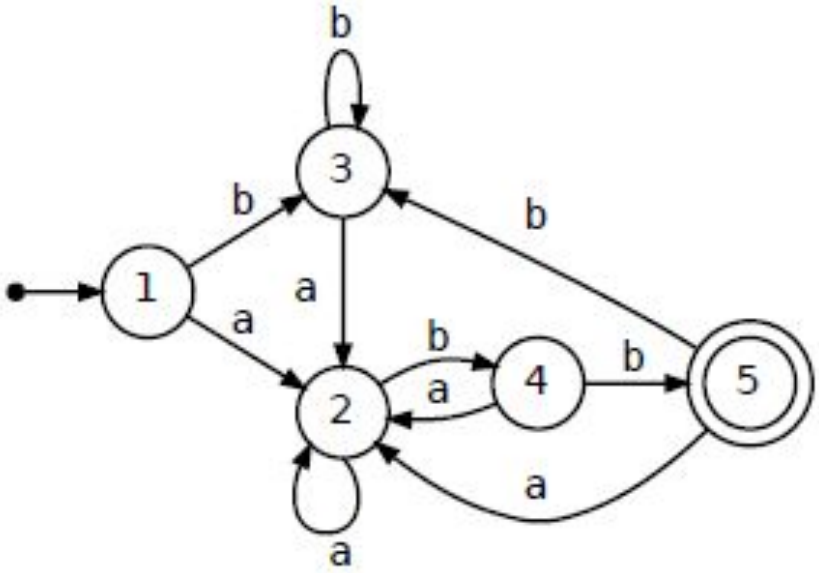
{1,4}

{2,3}

{2,4}

{3,4}

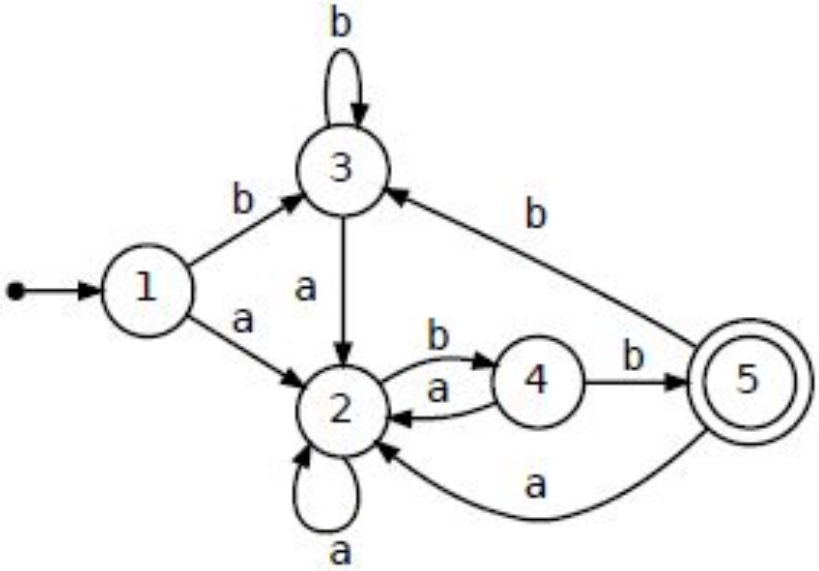
q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: **{1,2,3,4}** {5}
Step 2: {1,2

{1,2} => {{~~2,2~~},{3,4}}
{1,3}
{1,4}
{2,3}
{2,4}
{3,4}

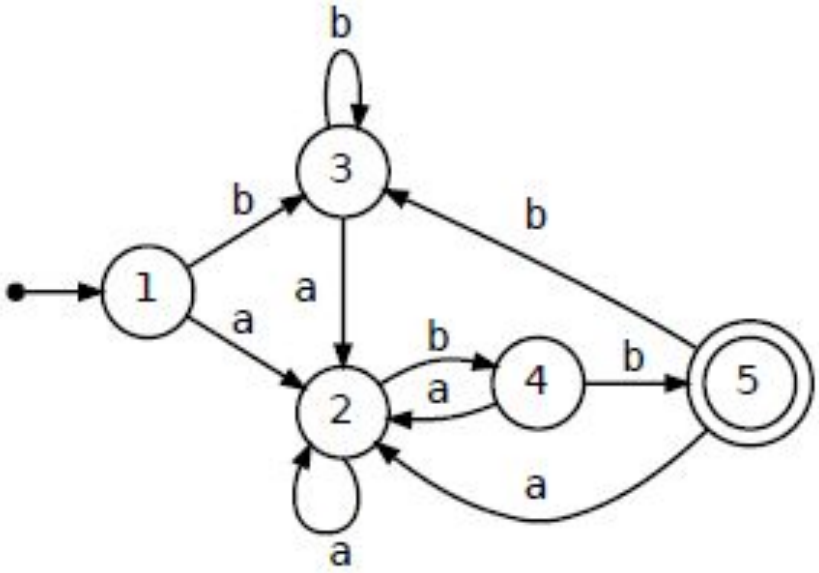
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⇒1	2	3
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3	2	3
4	2	5
5*	2	3



Step 1: **{1,2,3,4}** {5}
Step 2: {1,2

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{1,4}
{2,3}
{2,4}
{3,4}

q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: **{1,2,3,4}** {5}

Step 2: {1,2,3

{1,2} => {{~~2,2~~},{3,4}}

{1,3} => {{~~2,2~~},{~~3,3~~}}

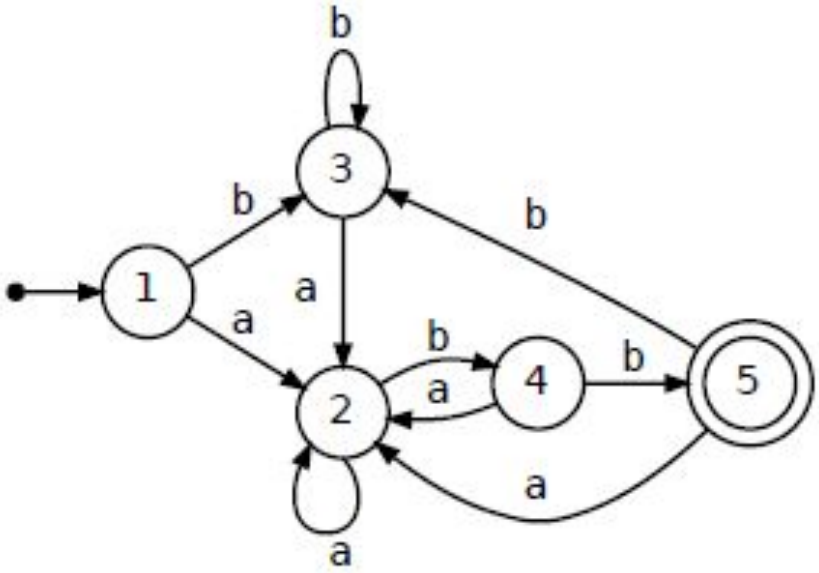
{1,4}

{2,3}

{2,4}

{3,4}

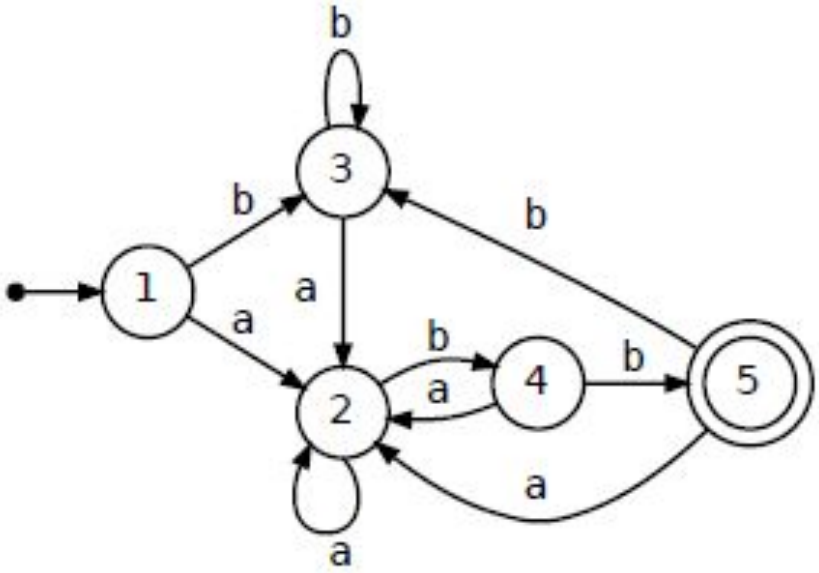
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4	2	5
5*	2	3



Step 1: **{1,2,3,4}** {5}
Step 2: {1,2,3

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{1,4} => {{~~2,2~~},{3,5}}
{2,3}
{2,4}
{3,4}

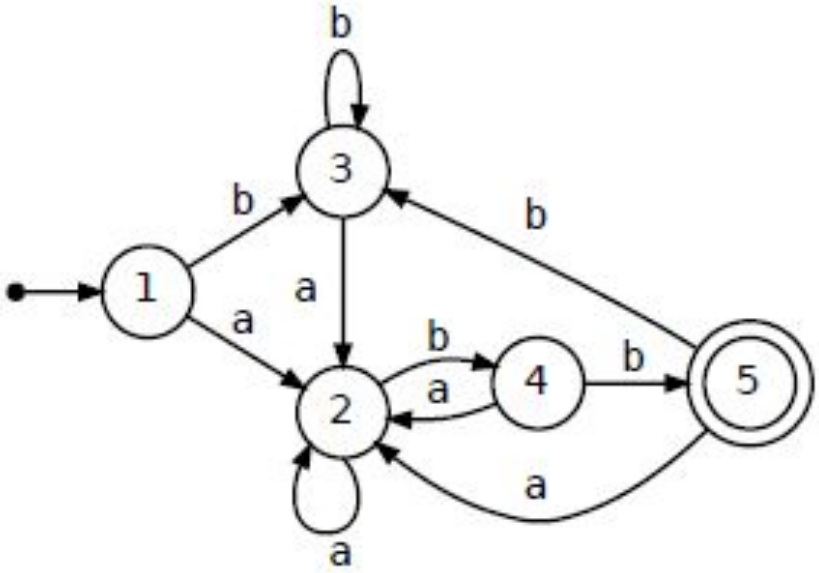
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Step 1: **{1,2,3,4}** {5}
Step 2: {1,2,3} {4} {5}

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{1,4} => {{~~2,2~~},{3,5}}
{2,3}
{2,4}
{3,4}

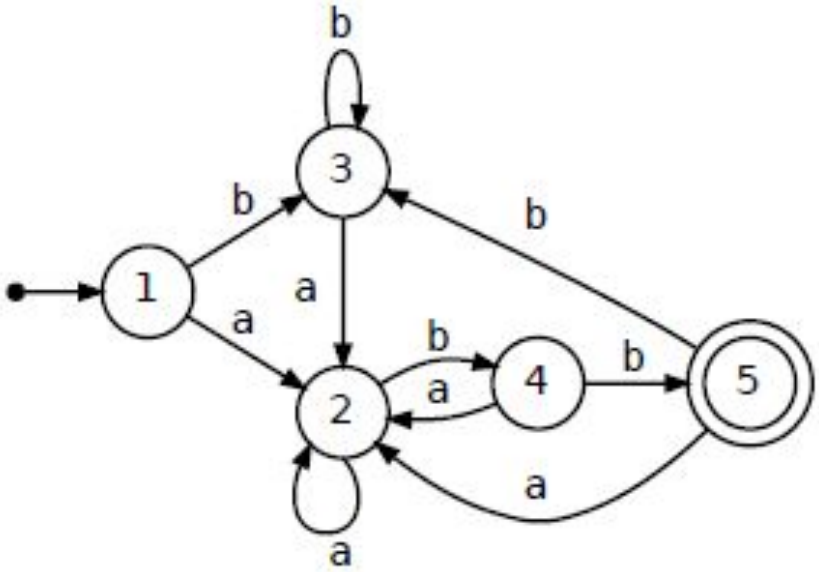
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Step 1: **{1,2,3,4}** {5}
Step 2: {1,2,3} {4} {5}

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{1,4} => {{~~2,2~~},{3,5}}
{2,3} ⇐⇒ {1,2} & {1,3}
~~{2,4}~~
~~{3,4}~~

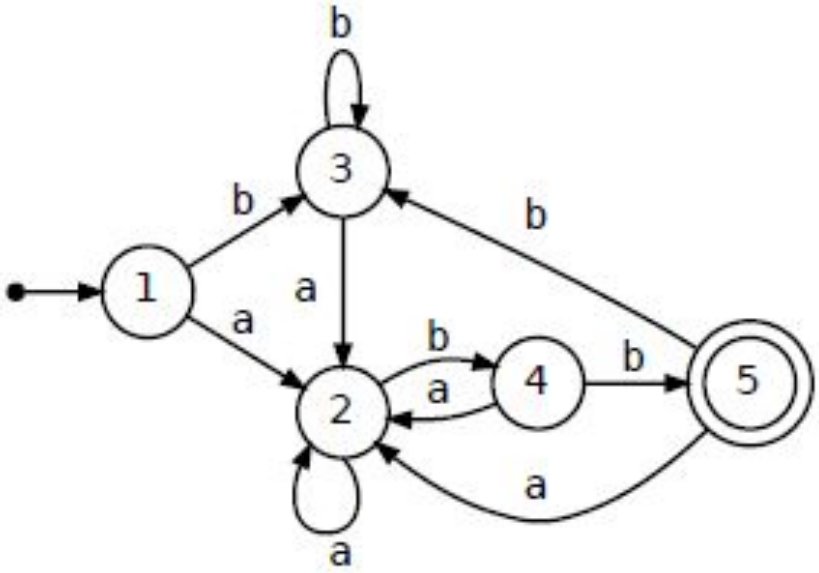
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Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3:

{1,2}
{1,3}
{2,3}

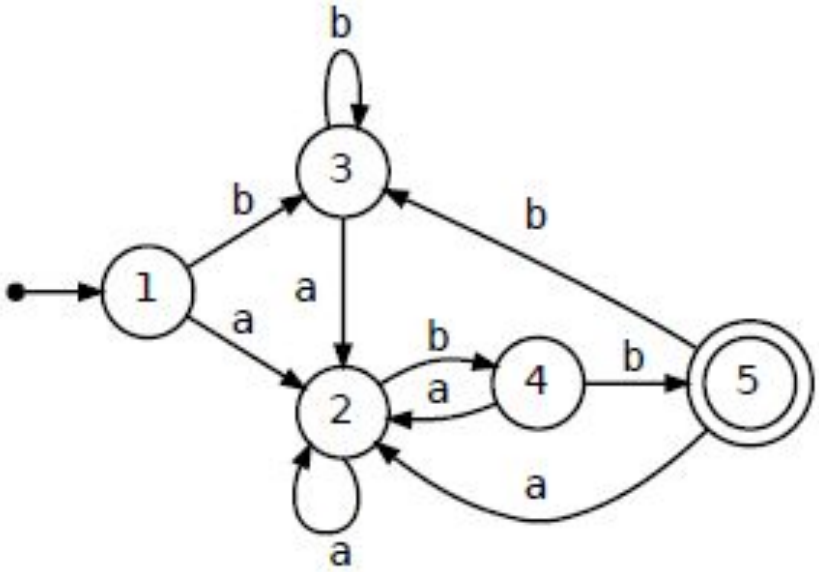
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Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3:

{1,2} => {{~~2,2~~},{3,4}}
{1,3}
{2,3}

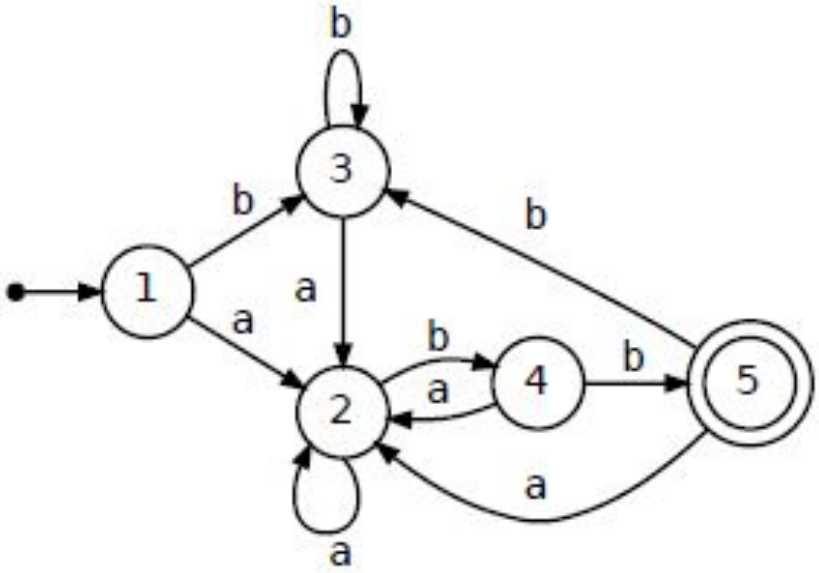
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4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3: {1 {2

{1,2} => {{~~2,2~~},{3,4}}
{1,3}
{2,3}

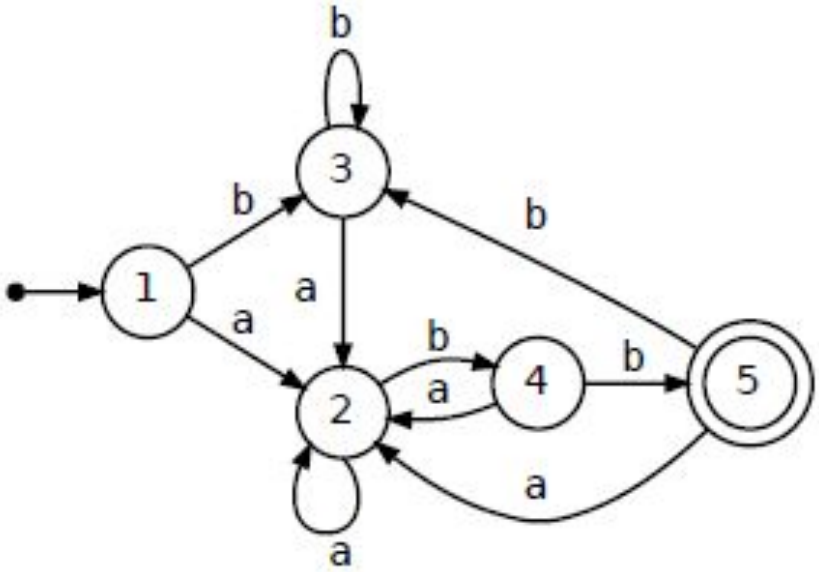
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Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3: {1 {2

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{2,3}

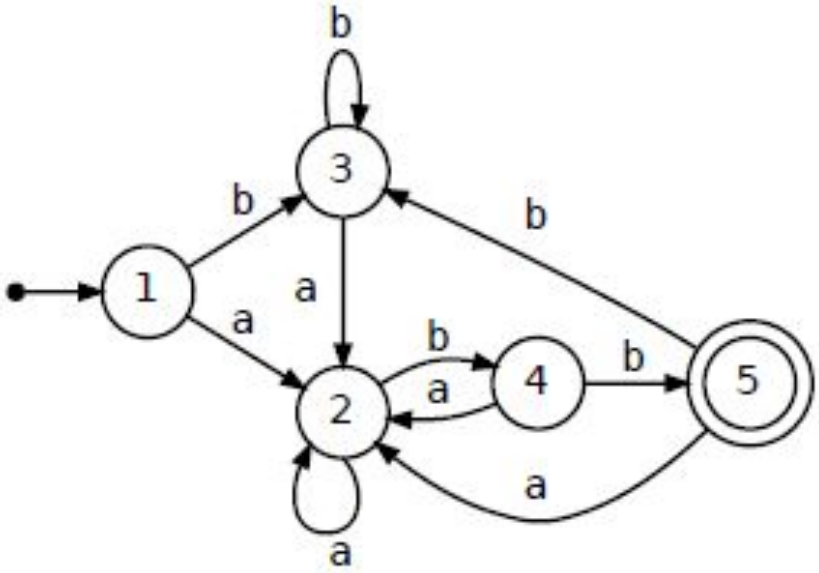
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3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3: {1,3 {2

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
{2,3}

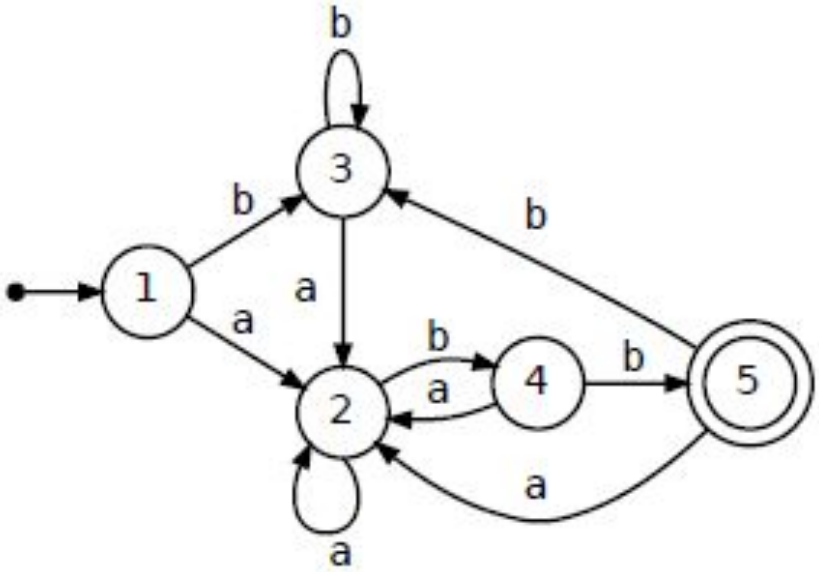
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⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3: {1,3} {2}

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
~~**{2,3}**~~

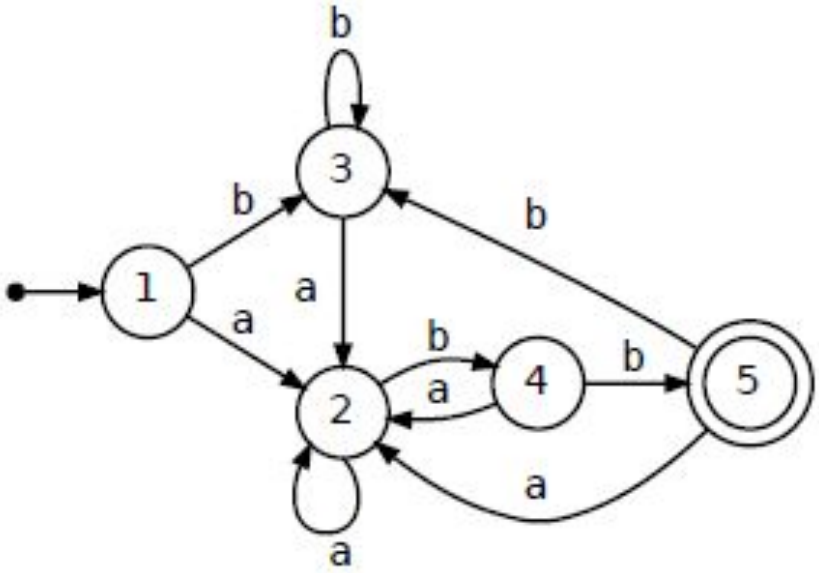
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⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: **{1,2,3}** {4} {5}
Step 3: {1,3} {2} {4} {5}

{1,2} => {{~~2,2~~},{3,4}}
{1,3} => {{~~2,2~~},{~~3,3~~}}
~~{2,3}~~

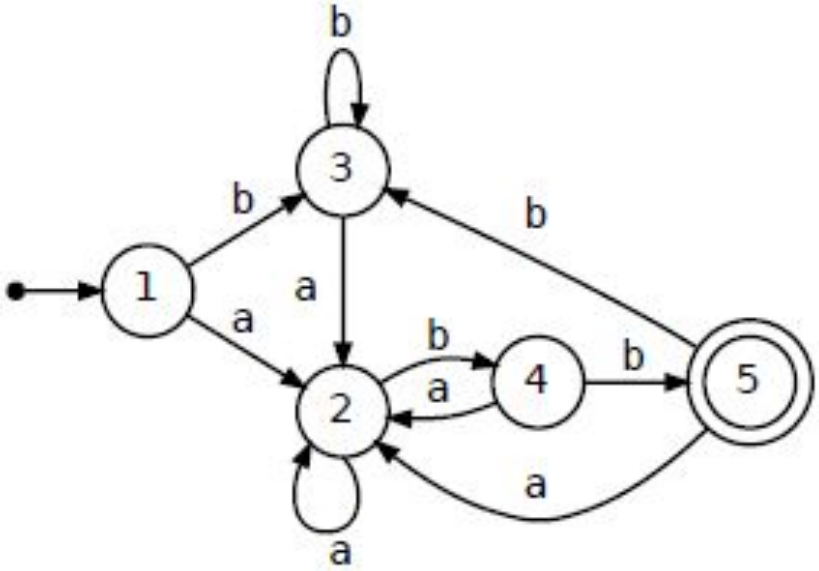
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⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: {1,2,3} {4} {5}
Step 3: **{1,3}** {2} {4} {5}
Step 4:

{1,3}

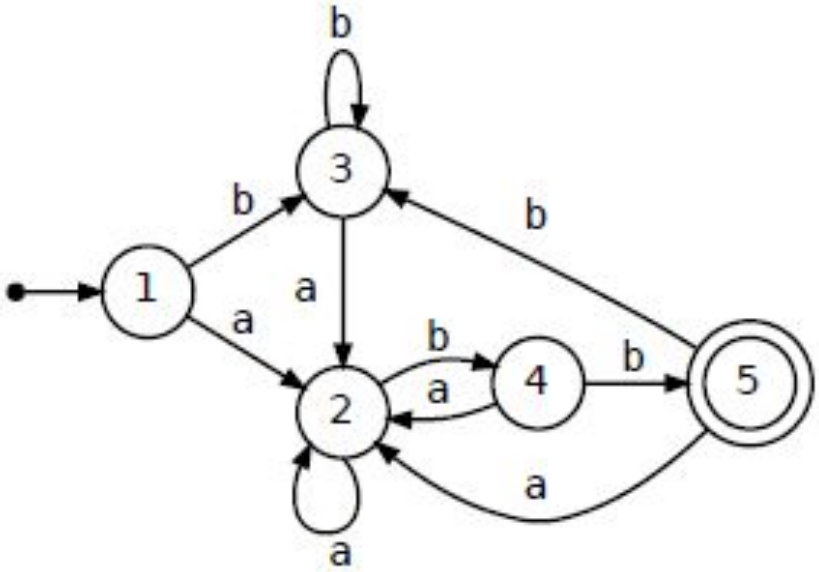
q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: {1,2,3} {4} {5}
Step 3: **{1,3}** {2} {4} {5}
Step 4:

{1,3} => {{~~2,2~~},{~~3,3~~}}

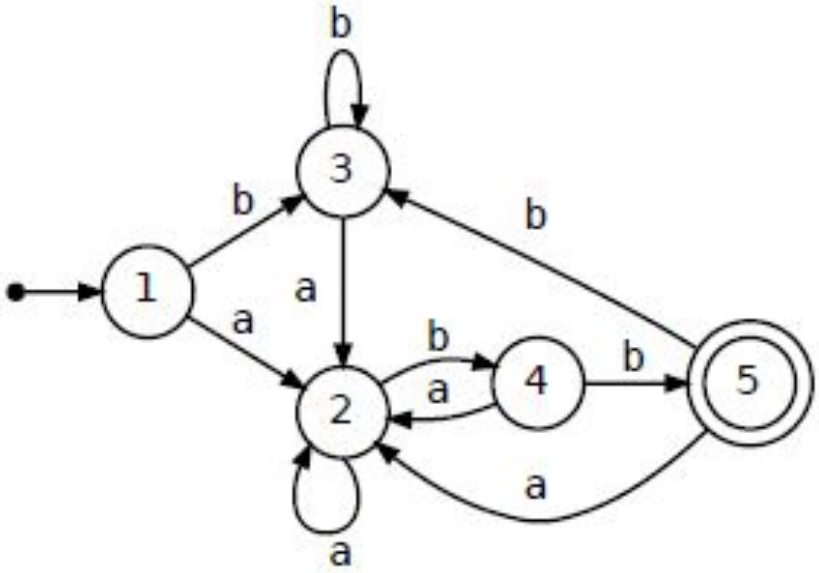
q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: {1,2,3} {4} {5}
Step 3: **{1,3}** {2} {4} {5}
Step 4: {1,3

{1,3} => {{~~2,2~~},{~~3,3~~}}

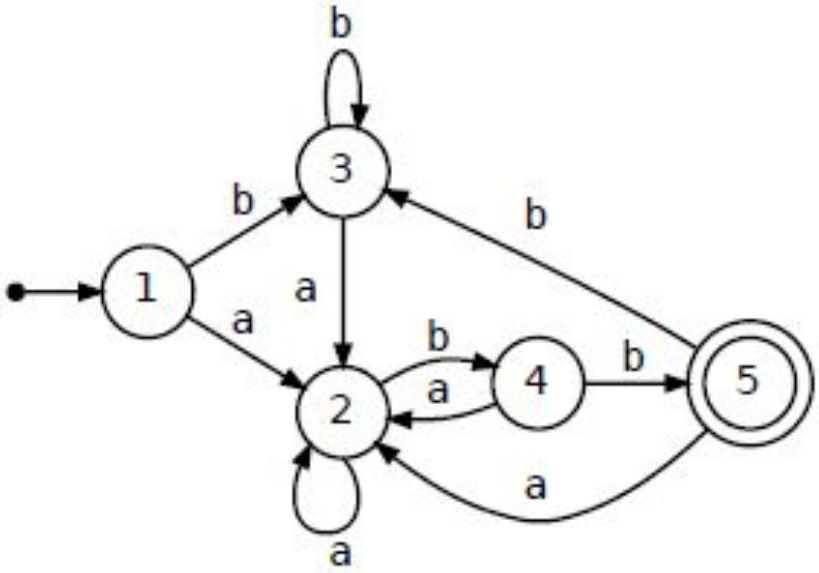
q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



Step 1: {1,2,3,4} {5}
Step 2: {1,2,3} {4} {5}
Step 3: **{1,3}** {2} {4} {5}
Step 4: {1,3} {2} {4} {5}

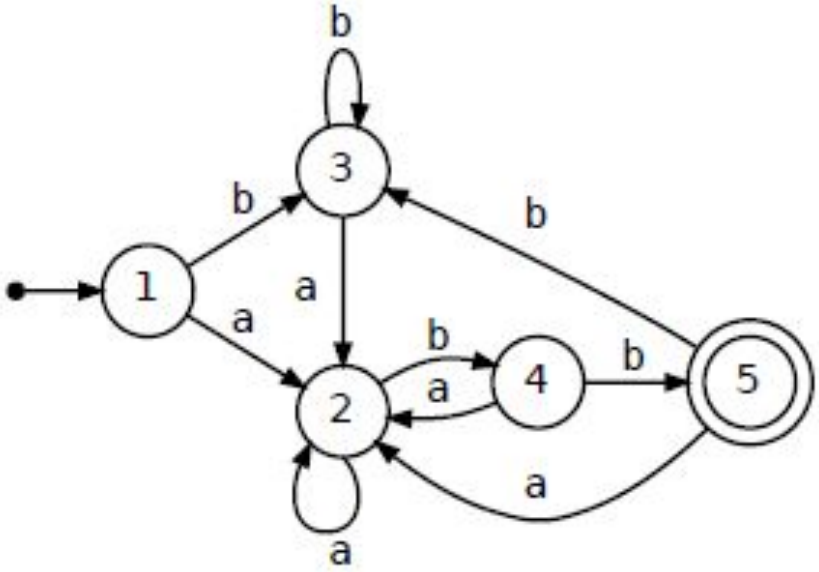
{1,3} => {{~~2,2~~},{~~3,3~~}}

q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3

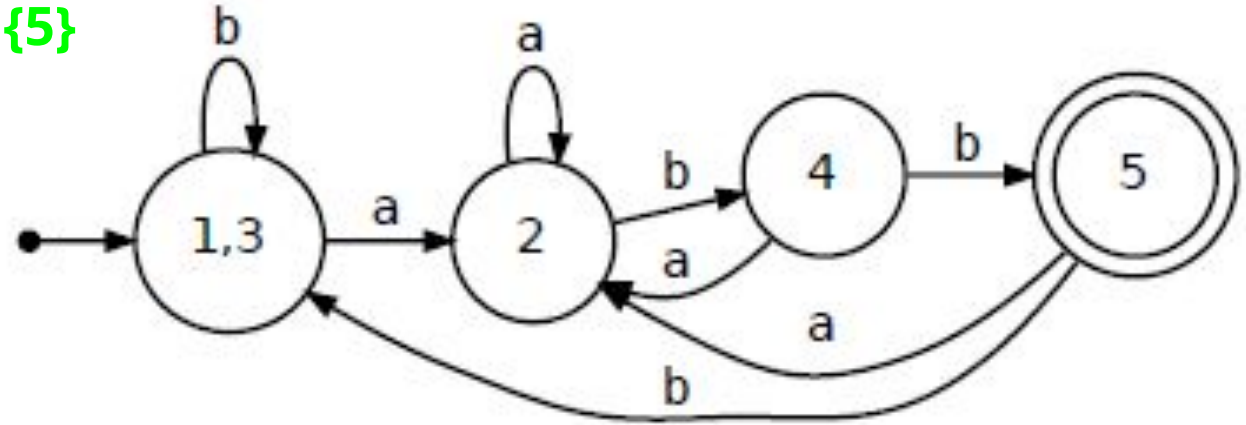


Step 1: {1,2,3,4} {5}
Step 2: {1,2,3} {4} {5}
Step 3: {1,3} {2} {4} {5}
Step 4: {1,3} {2} {4} {5}

q	δ(q,a)	δ(q,b)
⇒1	2	3
2	2	4
3	2	3
4	2	5
5*	2	3



{1,3} {2} {4} {5}



Example2

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

Step 1: **{0,1,3,4,5,6,7}** {2}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

Step 1: **{0,1,3,4,5,6,7}** {2}

Step 2:

{0,1}
{0,3}
{0,4}
{0,5}
{0,6}
{0,7}
{1,3}
{1,4}
{1,5}
{1,6}
{1,7}
{3,4}
{3,5}
{3,6}
{3,7}

{4,5}
{4,6}
{4,7}
{5,6}
{5,7}
{6,7}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{0,1} => {{1,6},{5,2}}
{0,3} => {{1,2},{5,6}}
{0,4} => {{1,7},~~{5,5}~~}
{0,5} => {{1,2},{5,6}}
{0,6} => {{1,6},{5,4}}
{0,7} => {{1,6},{5,2}}

{1,3}
{1,4}
{1,5}
{1,6}
{1,7}
{3,4}
{3,5}
{3,6}
{3,7}
{4,5}
{4,6}
{4,7}
{5,6}
{5,7}
{6,7}

Step 1: **{0,1,3,4,5,6,7}** {2}
Step 2: {0,4,6} {1} {3} {5} {7} {2}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

Step 1: **{0,1,3,4,5,6,7}** {2}
Step 2: {0,4,6} {1,7} {3,5} {2}

{0,1} => {{1,6},{5,2}}
{0,3} => {{1,2},{5,6}}
{0,4} => {{1,7},~~{5,5}~~}
{0,5} => {{1,2},{5,6}}
{0,6} => {{1,6},{5,4}}
{0,7} => {{1,6},{5,2}}
{1,3} => {{6,2},{2,6}}
~~{1,4}~~
{1,5} => {{6,2},{2,6}}
~~{1,6}~~
{1,7} => {{~~6,6~~},{~~2,2~~}}
~~{3,4}~~
{3,5} => {{~~2,2~~},{~~6,6~~}}
~~{3,6}~~
{3,7} => {{2,6},{6,2}}

~~{4,5}~~
~~{4,6}~~ ⇐⇒ {0,4}&{0,6}
~~{4,7}~~
~~{5,6}~~
{5,7} => {{2,6},{6,2}}
~~{6,7}~~

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{0,4}
{0,6}
{4,6}

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: **{0,4,6}** {1,7} {3,5} {2}
Step 3:

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{0,4} => {{1,7},~~{5,5}~~}
{0,6} => {{1,6},{5,4}}
~~{4,6}~~

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: **{0,4,6}** {1,7} {3,5} {2}
Step 3: {0,4} {6}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{1,7}

{3,5}

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: {0,4,6} **{1,7}** **{3,5}** {2}
Step 3: {0,4} {6}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{1,7} => {{~~6,6~~},{~~2,2~~}}

{3,5} => {{~~2,2~~},{~~6,6~~}}

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: {0,4,6} **{1,7}** **{3,5}** {2}
Step 3: {0,4} {6} {1,7} {3,5} {2}

q	δ(q,a)	δ(q,b)	
⇒0	1	5	{0,4}
1	6	2	
2*	0	2	{1,7}
3	2	6	
4	7	5	{3,5}
5	2	6	
6	6	4	
7	6	2	

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: {0,4,6} {1,7} {3,5} {2}
Step 3: **{0,4}** {6} **{1,7}** **{3,5}** {2}
Step 4:

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{0,4} => {{1,7},~~{5,5}~~}

{1,7} => {~~{6,6}~~,~~{2,2}~~}

{3,5} => {~~{2,2}~~,~~{6,6}~~}

Step 1: {0,1,3,4,5,6,7} {2}
Step 2: {0,4,6} {1,7} {3,5} {2}
Step 3: **{0,4}** {6} **{1,7}** **{3,5}** {2}
Step 4: {0,4} {6} {1,7} {3,5} {2}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

Step 1: {0,1,3,4,5,6,7} {2}

Step 2: {0,4,6} {1,7} {3,5} {2}

Step 3: {0,4} {6} {1,7} {3,5} {2}

Step 4: {0,4} {6} {1,7} {3,5} {2}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

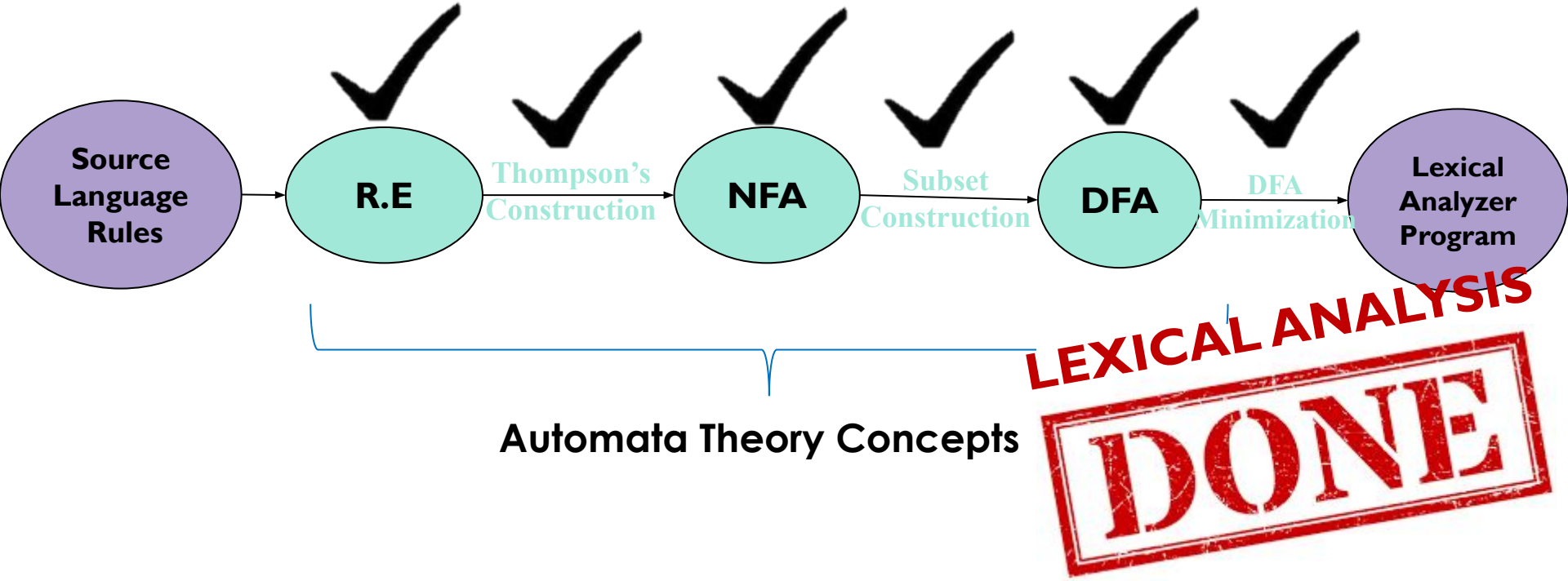
{0,4} {6} {1,7} {3,5} {2}

q	δ(q,a)	δ(q,b)
⇒0	1	5
1	6	2
2*	0	2
3	2	6
4	7	5
5	2	6
6	6	4
7	6	2

{0,4} {6} {1,7} {3,5} {2}

q	δ(q,a)	δ(q,b)
⇒0,4	1,7	3,5
6	6	0,4
1,7	6	2
3,5	2	6
2*	0,4	2

Lexical Analyzer and Automata




جامعة عين شمس
AIN SHAMS UNIVERSITY
كلية الحاسبات والمعلومات
Faculty of Computer & Information Sciences

Questions

List the tokens

```
while (x>0)
{
z=x+10;
x--;
}
```

Another Question:
Given the following tokens, use all of them to create a valid C++ code fragment.

a) Write a regular expression describing all strings over **Σ ={a ,b}** that starts with at least one **a** followed by zero or more sequences of **abb** and ends with zero or one **a.** [15 marks]

Note: *You should verify your answer with examples of* ***two accepted and two rejected strings.***



For the following Regular Expression:

a|abc*

- **a)** Construct the NFA using Thompson's Construction. *(5 marks)*
- **b)** Convert the NFA to DFA using Subset Construction. *(5 marks)*
- **c)** Minimize the resulting DFA using Hopcroft Algorithm. *(5 marks)*
- **d)** Determine **Three** accepted strings and **Three** rejected ones by the DFA. *(5 marks)*

Note: Show all Transition Tables required to solve (b) ,and (c)